

Question	Answer	Mark	Mark scheme	Additional guidance
23	No (supported)	P1 for a first step, eg to find the gradient of the line (normal) joining (0, 0) to (−3, 4), eg $\frac{4-0}{-3-0}$ ($= -\frac{4}{3}$) OR finds the length² for one side of triangle, eg $(-3 - -8)^2 + (4 - 0)^2$ ($= 41$) or $(-3 - 0)^2 + (4 - 0)^2$ ($= 25$) or 64 P1 for a second step, eg to find the gradient of the tangent, eg $-1 \div \text{“} -\frac{4}{3} \text{”}$ ($= \frac{3}{4}$) OR finds the length² for two sides of triangle, two of 25, 41 and 64 P1 for a third step, eg to find the gradient of the line joining (−8, 0) to (−3, 4), $\frac{4-0}{-3--8}$ ($= \frac{4}{5}$) or finds the equation of the tangent, eg $y = \frac{3}{4}x + \frac{25}{4}$ or $y - 4 = \frac{3}{4}(x - -3)$ OR for process to use Pythagoras’ rule, eg $41 + 25$ ($= 66$) or $64 - 25$ ($= 39$) or $64 - 41$ ($= 23$) C1 for No from correct figures and a complete process from comparison of gradients, eg $\frac{3}{4}$ and $\frac{4}{5}$ or showing the equation of the tangent does not pass through (−8, 0), eg when $x = -8$, $y = \frac{1}{4}$ (not 0) or when $y = 0$, $x = -\frac{25}{3}$ (not −8) OR correct figures from Pythagoras’ rule, eg $41 + 25$ ($= 66$) $\neq 64$ or $64 - 41 \neq 25$ or $64 - 25 \neq 41$ oe	Alternative processes may be seen and should be duly credited. Award 0 marks for No without complete and correct supportive working	